

“This Could Be The Day I Die”: Unpacking Interpersonal and Systems Trust in a Local Sharing Economy Community

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Figure 1: The Smart Box lockers at a train station used for sharing items

ABSTRACT

The rapid development of the “sharing economy” enables the effective and efficient coordination, acquisition, distribution, and sharing of many kinds of different resources. Beyond the well-known sharing economy services such as Airbnb and Uber, an increasing number of local sharing initiatives have established online platforms and services to facilitate access to the shared resources within their local communities. With the automation and complexity of digital tools and platforms, and the specific challenges of online sharing communities, users’ trust and reliance become increasingly critical for successful use and adoption. In our qualitative study in collaboration with two industry partners (a local sharing community and a large infrastructure provider in Switzerland), we unpack various perspectives of interpersonal trust in the community and the systems trust of the supporting technologies. On this basis, we elicited a set of design opportunities for future platforms in the context of sharing economy.

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CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

sharing economy, interpersonal trust, digital platforms, systems trust, qualitative study

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1 BACKGROUND

The “sharing economy” [4, 33] has become an umbrella term for a variety of social, economic, and business models that rely on networked technologies to enable access to shared resources. Popular sharing economy services (e.g., Airbnb, Uber) generate economic activity by matching providers and consumers to optimize resource use (e.g., apartments, vehicles). Some of these services merely facilitate the transaction and provide little or no support for social interaction and, hence, do not contribute substantially to social cohesion [13]. Some are even associated with negative social impacts

(e.g., discrimination [8], gentrification [38], precarious employment [37], monetization of personal data [41]).

In contrast, an increasing number of grassroots initiatives and member-driven organizations offer collections of shared items (e.g., books, tools, surplus food) and resources (e.g., community-owned spaces, equipment at fab labs, fertile land) intended to benefit local communities (see, e.g., [1, 11, 23, 29]). Many of these Community-Oriented Sharing Economy (COSE) initiatives provide network-driven technologies for shared resource management, communication, and coordination among members. Aligning with the UN's Sustainable Development Goals [14], these COSEs advocate environmental and social sustainability, promoting fairer distribution of goods and labor, resulting in a stronger sense of community and value for society [41].

However, these initiatives have different problems to solve than their global commercial counterparts. Prior research has identified interpersonal and organizational challenges that inhibit the development of COSEs [9, 12]. Some challenges stem directly from digitalization and can be approached from design and technology perspectives, specifically, issues of trust within the membership and supporting digital platforms [6, 23].

Trust is the basis for many human interactions and relationships. Furthermore, trust can be transferred to institutions, organizations, and technologies [2, 3]. With the automation and complexity of digital technologies, trust is especially important as it is a prerequisite for successful technology use and adoption [15, 20]. The importance of trust in technologies has been examined in different settings such as virtual communities [27], e-commerce [16, 35], online exchange communities [22], and other digital services that facilitate economic interactions among peers [3]. Interactive systems can have two roles in trust relationships [35]. The first is the mediator role in which a system mediates human interaction. In this role, *interpersonal* trust should be established. The second is the trustee role in which a system is the agency with which the human interacts. In this role, *systems* trust must be established.

In most cases, today's digital solutions are 'black boxes' [32] for users. Explaining how a digital solution is developed and how its algorithms function does not necessarily lead to more trust [32]. Therefore, new approaches are needed to engender trust and design trustworthy digital solutions. For example, trust-supporting design elements [26, 40] - platform features that establish end-user trust - can help establish trust in 'faceless' algorithms.

In online sharing communities, designers of platforms have the challenge of designing for both interpersonal and systems trust to ensure effective and meaningful interactions among their users [19, 36]. With the advent of rapid digitalization, the emergence of new digital solutions, and the pressure to stay competitive, service providers continually introduce new features and evolve existing mechanisms (at times through ethically-dubious deception techniques like dark patterns [18]), potentially affecting previously established trust in systems.

In our empirical study, we unpack elements of interpersonal and systems trust in the context of a local online sharing community which recently introduced a new semi-automated mechanism for resource exchange. Specifically, our research aims are: (i) to investigate the existing perceptions of interpersonal and systems trust that are prevalent in the local sharing economy community; and

(ii) to explore the design opportunities to engender trust within the local sharing economy community and toward digital platforms.

Our contribution is twofold. First, we provide an in-depth account of members' understandings of interpersonal and systems trust in a sharing community in Switzerland. Second, we propose a set of design opportunities to imbue trust within its membership and the supporting digital platforms, offering insights into how trust-supporting design elements can be integrated into technology-mediated exchanges.

2 CASE DESCRIPTION

This study is conducted in collaboration with two companies in Switzerland: a two-sided online marketplace [22] for household goods, Sharely¹, and a large infrastructure provider, the Swiss Federal Railways (SBB). Sharely aims to increase sharing and use of personal items (e.g., a drill, a bike pump, sports equipment), by advocating conscientious and sustainable resource consumption practices (e.g., "better to share than to buy"). Community members can post information about the items they have on Sharely's website or mobile app. These items can be borrowed from the lender for a small fee for a fixed period of time (e.g., a day, a week). Sharely takes a percentage of each transaction, and offers liability insurance (through a third-party company) in case of damage or loss. The pick-up and return of the items are decided by the lender and borrower with the help of the platform and the exchange of the item usually happens face-to-face.

Recently, Sharely added a new service that offers members an option for indirect resource exchange [24], i.e., in which pick-up and return can be done with no in-person contact. In a partnership with SBB, Sharely allows people to rent items using SBB Smart Boxes, newly repurposed luggage lockers at train stations (see Fig.1). The SBB Smart Boxes are Internet-enabled lockers that can be opened via a mobile app. In this context, Sharely provides a small set of its most popular items (e.g., a cordless drill, a high-pressure cleaner, a drone), which users can borrow via the Smart Boxes. Once the user requests an item, Sharely confirms its availability and places it in a Smart Box. Users can then retrieve it from the box (at the time of the writing, the service was in its pilot phase and offered only at one station in Zurich). The user then receives a notification on their mobile phone to open the locker to retrieve the item (Fig.2). After the lending period, the users return the item to the Smart Box and lock it using the mobile app.

The new system introduced by Sharely and SBB faces some trust-related challenges in regard to both interpersonal and systems trust [6]. Given the impersonal resource-sharing setting, interpersonal trust within the community could be hard to maintain. When such interactions replace traditional community-based peer-to-peer sharing, both the lender and the borrower may miss out on the opportunity to meet face-to-face as the trust is placed in a system rather than in other members [22]. Prior research indicates that face-to-face exchanges are critical for establishing trust in local resource-sharing communities [10]. Furthermore, building interpersonal trust in sharing environments is necessary to ensure the success of the sharing community, and to foster collaboration and social cohesion [28].

¹<https://www.sharely.ch>

PickUp & DropOff Open the Box.

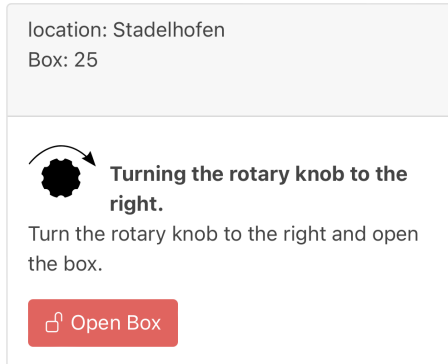


Figure 2: UI to open a locker

When it comes to systems trust, luggage lockers with smart locks are introduced as new technology to the community. systems trust improves the experience while using technology [31] and is also important for the adoption of new technology [20]. Lee and colleagues [25] argued that trust is based on what is known and rarely on what is unknown or new. They further elaborate that known technology can be used as a mediator of trust in the unknown. Therefore, successfully mediating trust from Sharely's website and service in this new sharing arrangement may pose yet another challenge to trust accrual in the community and their supporting technologies. More specifically, it is unknown how the introduction of this indirect resource exchange option (via SBB Smart Boxes) shapes members' perceptions of usability and reliability of the Sharely supporting technologies and how it affects people's attitudes toward privacy and safety, which have been regarded as constitutive properties of trust(worthiness) in computing systems [39].

3 METHOD

We collected empirical data (through an online questionnaire and interviews) about trust between the users of the platform and in the platform itself. We conducted interviews with members of the Sharely community to gain an understanding of their perspectives and concerns related to trust regarding direct and indirect exchanges, and their experiences with and opinions of the supporting digital tools. The interviews and the questionnaire give insight into the real-life experiences of the users and their reflections on trust. Specifically, our data sources included:

- (1) A short **questionnaire**, distributed through Sharely to their users, to collect basic demographic information and information about their experiences with Sharely (e.g., how many exchanges they have had), and attitudes toward the new Smart Box sharing option. We used this questionnaire to identify potential participants for the interviews.
- (2) In-depth online **interviews** with 14 users. The researcher explained the goal of the study and conducted a semi-structured interview about the user's experiences (both positive and

negative) with Sharely service and the Smart Box (if any) to create a rich qualitative account of participant's sharing interactions and views on interpersonal and systems trust.

- (3) **Contextual** information such as users' public profiles on the platform, the items they share, and reviews of their interactions with other users. The contextual information provided additional perspectives on interactions on the platform.

We received 67 responses to the questionnaire from members who had engaged in direct and indirect exchanges with Sharely. We invited 14 participants for follow-up interviews. The average age of the interview participants was 44.8 (SD=9.4), four female, eight male, two non-binary. Our study participants held various occupations, including marketing specialist, logistics manager, social worker and deacon. Table 1 provides an overview of our interviewees.

Table 1: Overview of interviewees.

No.	Pseudonym	Age	Gender	Occupation
1	Anna	43	Female	Product Manager
2	Paul	38	Male	Finance
3	Sophie	65	Non-binary	Homemaker
4	Ben	43	Male	Business Analyst
5	Mia	38	Female	Teacher
6	Jonas	38	Male	Deacon
7	Henry	51	Male	Social Worker
8	Finn	40	Non-Binary	IT Specialist
9	Felix	36	Male	Self-Employed/Educator
10	Lea	34	Female	Project Manager
11	Noah	38	Male	Logistics Manager
12	Luis	46	Male	IT Specialist
13	Emma	47	Female	Homemaker
14	Max	63	Male	Marketing

To ensure that we covered both sharing options (i.e., direct and indirect exchanges), we recruited Sharely members who have participated in the new sharing setup at the train station, and members who have not yet done so. We asked participants to describe their memorable positive and negative sharing experiences and invited them to reflect on challenges they had experienced. We also discussed the items they have shared by browsing their profile and history of exchanges on the Sharely website. Additionally, to learn more about trust in the community, we recruited a few non-active members of Sharely, i.e., registered members who have not engaged in lending/borrowing transactions, and asked for reflections on why this might be the case. Finally, we created screenshots of two fictional user profiles with largely positive and slightly negative reviews (Fig. 3) and discussed their willingness to rent with them and their perceived trustworthiness. As an incentive, we offered participants a gift voucher worth approximately \$25 USD for their time and effort. We conducted our interviews online using an online conferencing tool (Zoom). Interviews were between 45 and 60 minutes long. Interviews were audio-recorded for transcription.

We used a deductive and inductive coding strategy from grounded theory [17] and analyzed the materials thematically using the reflexive thematic analysis technique [5]. The goal of the analysis was to draw insights from the the participants' experiences and

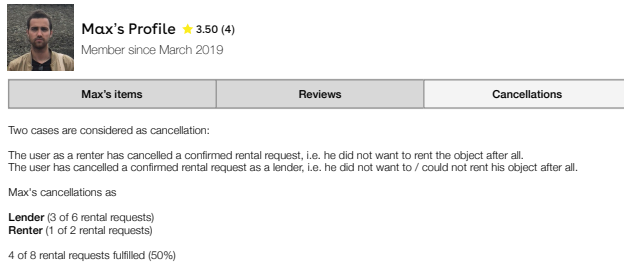


Figure 3: Fictional user profile used in our interviews

derive empirical categories to describe the phenomena observed. The empirical categories are “grounded” in the statements of the participants. We held meetings among the researchers team to interpret the findings and discuss differences and overlaps between the categories until we reached a consensus for each theme. Following the synthesis of Wing [39] on trustworthy computing, we reconstructed our analysis according to their suggested trust facets: reliability, privacy, safety and usability. We traveled between raw data, codes, and themes to critically explore semantic and latent relationships in our data. These relationships were identified inductively and continued to evolve as we wrote up the findings.

4 FINDINGS

We present examples that capture the participants’ perceptions of interpersonal and systems trust and support them with quotes from the interviews and questionnaire. We have formulated these perceptions as tensions in the Sharely community or as needs regarding trust in the community and the supporting technologies. We refer to participants by pseudonyms to protect their privacy.

4.1 Interpersonal Trust

4.1.1 Dichotomy in interpersonal trust. There is an inherent sense of trust in the Sharely community: like-minded individuals are eager to lend their possessions to strangers with little hesitation, especially when support (i.e., liability insurance) is provided which reduces their risk in the event of damage or loss. Anna, 43, reflected: *“I don’t care at all who’s renting my stuff and I also wouldn’t read anything [on their profile] about them [...] I believe people who [use] Sharely have a similar mindset. I’m not attached to stuff [I’ve listed] here, but if it’s broken or lost I wouldn’t be [too disappointed], and I can replace it. So I fully trust [members] and if not, there is an insurance. [...] I trust insurance.”*

Ben, 43, explained that for some items, personal contact is crucial to determine the trustworthiness of an exchange partner: *“[...] with expensive items, it’s important for me that I have this personal contact. Like with the racing bike. That’s a good example. I ask, ‘Have you ever rode a racing bike?’ And when he answers yes, but when he comes and sits on the bike and makes a test round. Then I see ‘you never sat on a racing bike’, then I’d say ‘Hey, I don’t think that’s the right item for you.’ I don’t want to give my racing bike to someone who doesn’t have a clue.”* Anna and Ben’s reactions illustrate a dichotomy in the community between wanting personal contact to establish trust and not caring who the renter is.

4.1.2 Community-building vs. Convenience. We also observed a tension between the community-building priorities and the transactional nature of the platform. Henry, 51, who regularly rents a pair of crampons from another Sharely member, reflected on the social benefits of the community and the value of exchanging items in person: *“[Meeting face-to-face] also makes it more personal. If I’d got the crampons from a locker, I wouldn’t have got to know the other person. I wouldn’t have heard where she has been and what she’d recommend. That also contributes to the quality: you meet people with a common hobby, in best case directly nearby, directly in the neighborhood.”*

In contrast, Lea, 34, commented on the convenience of the Smart Box: *“It is very convenient that you can pick up the object at any time without having to make a hand-over appointment and be in [direct] contact with [any]one. [...] I would even prefer [to use the box].”* This tension suggests that with indirect exchanges, trust may shift from an individual to the company. Prior work has come to similar findings in the context of short-term rentals, where Airbnb users favor the convenience of faceless check-ins/outs using smart keys/locks rather than opt-in for more personal experiences with the presence of a host [22]. Lea further elaborated on the increased accountability of exchanges when trust transfers from an individual to the company: *“It sounds a bit strange, but for me, it was knowing that you rent something from Sharely directly, rather than a person, kind of gives you a bit more security [that] the rented product is in good shape. [...] Whereas with, with people, you always hope and it always worked well, but there’s always this thought, what if [...] the thing is not as described or something.”*

4.1.3 Unreliable behavior. When browsing through a fictional user profile with some negative reviews and a number of canceled item requests (Sharely provides this metric on users, see Fig.3), Jonas, 38 shared his opinion: *“I’d need more information on how these cancellations appeared. I had a cancellation of a child carrier last year shortly before Christmas. It has happened during the pandemic: [they] couldn’t come because of covid. So, I wasn’t mad that [we] didn’t rent [the item] out because of this. Even if this cancellation had arrived a day before the start of the rent, it’s not really predictable. So, I’d refrain from judging someone because they canceled.”* In contrast to Jonas, many of our participants expressed disappointment about last-minute cancellation, such as requests cancelled without further explanation, unanswered requests on the platform, no-shows, or items not ready at the scheduled pick-up time.

4.1.4 Concerns regarding personal safety. When prompted about safety concerns, Sophie, 65, summarized her experience: *“to be honest [it was] not [a concern] until I went to pick up the item, it turns out that I needed to go and pick it up at some ‘atelier’, that was in a back alley. I was like [...] this could be the day I die.”* Participants have reflected on the potential for uncomfortable situations, why they might arise, and how they could be addressed like in the case of Finn, 40: *“[I am] tall, bearded. It’s dark in the stairwell, and [when] I say: ‘[the item] is in the basement, come with me’ if I haven’t prepared it yet. I’ve noticed: not everyone is equally comfortable with that [...], especially women. Not all of them, but sometimes I have the impression that perhaps they are not just happy about following me into the cellar. I can avoid that by putting things beforehand at the entrance, or I [simply] say: ‘You can also wait.’ In the beginning, I didn’t realize*

that and then because of the [others'] behavior, I changed and told people: "I'll be right back."

4.2 Systems Trust

4.2.1 Uncertainty in technology-mediated sharing. Regarding systems trust, we have observed some challenges related to the transparency of the Smart Box. For example, Lea was unsure when she could open the locker, and whether she had to be physically at it: *"I got the link [to open the locker] when I was at home. I was very curious to see what would have happened if I clicked on OPEN now. So I'm not sure how the technology works, if it actually works only if you're in front of it or if it would have really opened [the locker] while I was at home. So I could imagine that, by mistake, people click on OPEN now. Would it open?! Would it have also locked again after some time [...] I was curious, but I didn't touch it [before I got to the station] That's maybe one thing that would add — another layer of security [when interacting with the box]."*

4.2.2 Usability concerns. Participants described several minor usability concerns about the functionality of the Sharely website (e.g., inadequate messaging interface, confusing site navigation and section naming, the lack of ability to reverse undesired actions), and the Smart Box. Participants found many small usability issues frustrating and were willing to provide feedback on how to improve the service. For example, Jon, 30 while responding to the questionnaire summarized a specific issue related to the proof of return: *"I am very satisfied [with the Smart box]; the simplicity of the service is impressive. [The] only negative comment: an indication of [making a] photo on return could be made more visible [in the UI]."*

4.2.3 Various privacy and safety needs. We discovered different privacy understandings and attitudes among our participants. For example, Luis, 46, who works in IT, was aware of what data he shared with Sharely and what the major concerns were: *"[I share] standard data: name, city, maybe date of birth, email address, cell phone number, and my items are in there with photos. I think not much more [...] of course [Sharely knows] how many rentals I have. That is also displayed now [on the profile page]. But that, [Sharely] want[s] to know how many [items I have], what's going on [...]. It [also] takes a lot of effort to protect data [...] if someone finds out my address and [Sharely] password and my date of birth, it's not the end of the world for me."* Conversely, Lea had trouble understanding the data exchange: *"I got the email, you clicked on the link and then kind of a button popped up saying open the [Smart] box now [...] but I don't know what type of data was shared between me and the box and how exactly that worked."*

Some participants also expressed concerns about the item's safety, which needed to be addressed in case this service were to be scaled up in the future, for example with multiple items in one locker. Lea explained: *"There was another drone in the same locker. I was wondering if this could be "risky" to leave such an expensive or generally several rented items in the same locker. It could be misused or a renter could say he only took the rented object."*

4.2.4 Service reliability. During our study, Sharely has made significant changes to the user interface on the website. In line with the prior art [34], many users were averse to change at the inception. However, this new website release affected mobile app users, as

the app became virtually unusable. Jonas explained: *"Until the new layout, everything worked smoothly. Maybe it also has to do with the app not working currently. That it's kind of tedious right now. Also, a friend who rented me the cheese set said that she doesn't receive the messages correctly or she doesn't really get notified that she has [new] messages. Eventually, I saw her message because she wrote [me] a text message. [...] if I would manage such a platform, then I would heavily check that it runs perfectly to 99% at the time it goes live."* This suggested that the service was unreliable and affected the experience of users during the service update.

5 DESIGN CONSIDERATIONS

We presented our participants' reflections on interpersonal trust within the Sharely community and the supporting technologies (i.e., the website, the app, and the Smart Box). The members' perceptions of interpersonal trust covered various tensions: the importance of face-to-face interactions and the convenience of indirect exchanges, reducing risks when interacting with strangers, reflections on the unreliability of peers' behaviors, and concerns regarding personal and items' safety. When it comes to systems trust, we illustrated the participants' diverse concerns of privacy and safety, issues regarding technology reliability and predictability, and multiple usability concerns. Based on these findings, we synthesized a set of design opportunities to engender trust within local sharing communities and their supporting technologies.

For interpersonal trust, we encourage designers of COSEs to:

- **Provide opportunities to facilitate reliable behavior.** Our participants were often frustrated when items requests or messages were not answered. We recommend platform designers nudge or incentivize responses to unread or unanswered messages. Contextual nudges from Gmail ² can be seen as an example of how this could be integrated.
- **Improve the feedback process.** Participants were disappointed by cancellation, but context can be an important factor in the experience. As Jonas expressed, cancellations are understandable for reasons such as illness. Therefore allowing users to specify a reason for cancellation would create more transparency in community interactions and potentially lead to greater satisfaction with the community.
- **Ensure a safe and trustworthy environment.** Meeting strangers for the first time can be daunting. Designers could offer options for more detailed profile descriptions and pictures to help introduce members and reduce "stranger status", thus increasing overall trust among peers. For example, peer similarity (e.g., common hobbies) [30], peer proximity [10], and safe places for exchanges can alleviate some safety concerns.

For the systems trust, we believe that the following design opportunities may facilitate trust:

- **Improve usability of pre- and post-sharing actions.** Tackling usability issues on the platform might seem straightforward; however, it can improve user experience with the service, especially in new sharing options, such as the Smart Box.

²<https://i0.wp.com/9to5google.com/wp-content/uploads/sites/4/2018/05/gmail-nudge.png>

- **Provide reliable service.** Change aversion [34] caused by abrupt changes in design created discomfort for our interviewees. Proactive communication about upcoming website changes and ensuring reliability of services during updates or maintenance are good practices to ensure a positive user experience.
- **Make the interaction with the service predictable.** When introducing new sharing routines (such as the Smart Box exchange), non-reversibility of actions on the website/app may damage the user's reputation on the platform (e.g., forgetting to close a locker may yield a drop in rating). We recommend designers make a service predictable using known UI patterns such as the Snackbar³ element used in Google Mail to undo undesired actions (e.g., due to slips).
- **Alleviate existing privacy concerns.** Designers could opt for transparency in transferring user data among intermediaries. We recommend designers consider contextual privacy labels [21] to let users know what information will be shared with third parties when using the service.

6 CONCLUSION AND FUTURE WORK

Through an in-depth qualitative study of a local sharing economy community, we have revealed various perspectives of interpersonal trust among members and the systems trust in digital technologies. We have elicited a set of design opportunities to help address trust issues through platform design and offered examples of trust-supporting design elements that can be employed in the context of peer exchanges.

In our future work, we aim to collect more data on indirect resource exchanges, as the infrastructure provider we worked with in this case study plans to expand their Smart Box service nationally. Further insights into members' sharing practices with Smart Boxes would improve the service experience. We see the potential that this could also inform the development and deployment of IoT automation in the context of sharing economy (e.g., the increasing use of smart locks to grant access to shared apartments [7]). To date, the decision about renting out a resource has been exclusively made by a host/owner of a resource, but we envision that an AI-powered agency could assume the role of a decision-maker in the near future, bringing up a new set of characteristics to determine systems trustworthiness such as fairness, accountability, and transparency. Future research would also benefit from the controlled experiments concerning the trustworthiness of design interventions based on the opportunities we have presented in this research. Ultimately, exploring trust-supporting design elements for future (AI-powered) sharing economy platforms and services marks a salient motivation for our future work.

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REFERENCES

- [1] Susanne Bødker, Henrik Korsgaard, and Joanna Saad-Sulonen. 2016. A Farmer, a Place and at least 20 Members- The Development of Artifact Ecologies in Volunteer-based Communities. In *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing - CSCW '16*. ACM Press, New York, New York, USA, 1140–1154. <https://doi.org/10.1145/2818048.2820029>
- [2] Balázs Bodó. 2021. Mediated trust: A theoretical framework to address the trustworthiness of technological trust mediators. *New Media & Society* 23, 9 (2021), 2668–2690.
- [3] Rachel Botsman. 2017. *Who can you trust?: how technology brought us together—and why it could drive us apart*. Penguin UK.
- [4] Rachel Botsman and Roo Rogers. 2010. *What's mine is yours. The rise of collaborative consumption*. Harper Collins, New York.
- [5] Virginia Braun and Victoria Clarke. 2021. *Thematic Analysis : A Practical Guide*. SAGE Publications Ltd, London. <http://digital.casalini.it/9781526417305http://digital.casalini.it/5282292>
- [6] Coye Cheshire. 2011. Online Trust, Trustworthiness, or Assurance? *Daedalus* 140, 4 (2011), 49–58. https://doi.org/10.1162/DAED_a_00114
- [7] Rajib Dey, Sayma Sultana, Afsaneh Razi, and Pamela J. Wisniewski. 2020. Exploring Smart Home Device Use by Airbnb Hosts. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (Honolulu, HI, USA) (CHI EA '20)*. Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3334480.3382900>
- [8] Benjamin G. Edelman and Michael Luca. 2014. Digital Discrimination: The Case of Airbnb.com. *SSRN Electronic Journal* (jan 2014). <https://doi.org/10.2139/ssrn.2377353>
- [9] Anton Fedosov, Airi Lampinen, Tawanna R. Dillahunt, Ann Light, and Coye Cheshire. 2019. Cooperativism and Human-Computer Interaction. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems - CHI EA '19*. ACM Press, New York, New York, USA, 1–4. <https://doi.org/10.1145/3290607.3311751>
- [10] Anton Fedosov, Airi Lampinen, William Odom, and Elaine M. Huang. 2021. A Dozen Stickers on a Mailbox: Physical Encounters and Digital Interactions in a Local Sharing Community. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW3 (jan 2021), 1–23. <https://doi.org/10.1145/3432939>
- [11] Anton Fedosov, William Odom, Marc Langheinrich, and Ron Wakkary. 2018. Roaming Objects: Encoding Digital Histories of Use into Shared Objects and Tools. In *Proceedings of the 2018 on Designing Interactive Systems Conference 2018 - DIS '18*. ACM Press, New York, New York, USA, 1141–1153. <https://doi.org/10.1145/3196709.3196722>
- [12] Anton Fedosov, Ozge Subasi, Lisa Ochsenbein, and Elaine Huang. 2022. Supporting Designers in the Sharing Economy Through a Generative Design Cards Toolkit. In *Creativity and Cognition*. ACM, New York, NY, USA, 498–504. <https://doi.org/10.1145/3527927.3535203>
- [13] Koen Frenken and Juliet Schor. 2017. Putting the sharing economy into perspective. *Environmental Innovation and Societal Transitions* 23 (2017), 3–10. <https://doi.org/10.1016/j.eist.2017.01.003>
- [14] Mayo Fuster Morell, Ricard Espelt, and Melissa Renau Cano. 2020. Sustainable Platform Economy: Connections with the Sustainable Development Goals. <https://doi.org/10.3390/su12187640>
- [15] David Gefen, Elena Karahanna, and Detmar W Straub. 2003. Trust and TAM in online shopping: An integrated model. *MIS quarterly* (2003), 51–90.
- [16] David Gefen and Detmar W Straub. 2004. Consumer trust in B2C e-Commerce and the importance of social presence: experiments in e-Products and e-Services. *Omega* 32, 6 (2004), 407–424.
- [17] Barney G Glaser and Anselm L Strauss. 2009. *The discovery of grounded theory: Strategies for qualitative research*. Transaction Publishers.
- [18] Colin M. Gray, Yubo Kou, Bryan Battles, Joseph Hoggatt, and Austin L. Toombs. 2018. The Dark (Patterns) Side of UX Design. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (Montreal QC, Canada) (CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3173574.3174108>
- [19] Florian Hawlitschek, Timm Teubner, Marc Thomas Philipp Adam, Nils S Borchers, Mareike Moehlmann, and Christof Weinhardt. 2016. Trust in the sharing economy: An experimental framework. (2016).
- [20] Axel Hoffmann, Holger Hoffmann, and Matthias Söllner. 2013. Fostering initial trust in applications—developing and evaluating requirement patterns for application websites. In *21st European Conference on Information Systems (ECIS), Utrecht, The Netherlands*.
- [21] Patrick Gage Kelley, Joanna Bresee, Lorrie Faith Cranor, and Robert W. Reeder. 2009. A "Nutrition Label" for Privacy. In *Proceedings of the 5th Symposium on Usable Privacy and Security (Mountain View, California, USA) (SOUPS '09)*.

³<https://uxdesign.cc/an-underrated-ux-gmail-undo-send-d2a1ff04251f>

- Association for Computing Machinery, New York, NY, USA, Article 4, 12 pages. <https://doi.org/10.1145/1572532.1572538>
- [22] Airi Lampinen and Barry Brown. 2017. Market Design for HCI. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems - CHI '17*. ACM Press, New York, New York, USA, 4331–4343. <https://doi.org/10.1145/3025453.3025515>
- [23] Airi Lampinen, Vilma Lehtinen, Coye Cheshire, and Emmi Suhonen. 2013. Indebtedness, reciprocity, and fairness in local online exchange. In *Proceedings of the 2013 conference on Computer supported cooperative work - CSCW '13*. ACM Press, New York, New York, USA, 661. <https://doi.org/10.1145/2441776.2441850>
- [24] Matthew V Law, Mor Naaman, and Nicola Dell. 2018. ShareBox: Designing A Physical System to Support Resource Exchange in Local Communities. In *Proceedings of the 2018 Designing Interactive Systems Conference*. ACM, New York, NY, USA, 1155–1167. <https://doi.org/10.1145/3196709.3196763>
- [25] Minha Lee, Lily Frank, and Wijnand IJsselstein. 2021. Brokerbot: A cryptocurrency chatbot in the social-technical gap of trust. *Computer Supported Cooperative Work (CSCW)* 30, 1 (2021), 79–117.
- [26] Jan Marco Leimeister, Winfried Ebner, and Helmut Krcmar. 2005. Design, implementation, and evaluation of trust-supporting components in virtual communities for patients. *Journal of Management Information Systems* 21, 4 (2005), 101–131.
- [27] Jan Marco Leimeister, Pascal Sidiras, and Helmut Krcmar. 2006. Exploring success factors of virtual communities: The perspectives of members and operators. *Journal of organizational computing and electronic commerce* 16, 3–4 (2006), 279–300.
- [28] Ann Light. 2022. Trust in Collaborative Economies and How to Study It: Relational Assets and the Making of More-than-Strangers. *Ethnographies of Collaborative Economies across Europe* (2022), 13.
- [29] Ann Light and Clodagh Miskelly. 2014. Design for sharing. *Northumbria University/The Sustainable Society Network* (2014).
- [30] Carol Moser, Paul Resnick, and Sarita Schoenebeck. 2017. Community Commerce: Facilitating Trust in Mom-to-Mom Sale Groups on Facebook. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 4344–4357. <https://doi.org/10.1145/3025453.3025550>
- [31] Bonnie M Muir. 1987. Trust between humans and machines, and the design of decision aids. *International journal of man-machine studies* 27, 5–6 (1987), 527–539.
- [32] Wolter Pieters. 2011. Explanation and trust: what to tell the user in security and AI? *Ethics and information technology* 13, 1 (2011), 53–64.
- [33] Juliet Schor. 2016. Debating the sharing economy. *Journal of Self-Governance and Management Economics* 4, 3 (2016), 7–22.
- [34] Aaron Sedley and Hendrik Müller. 2013. Minimizing Change Aversion for the Google Drive Launch. In *CHI '13 Extended Abstracts on Human Factors in Computing Systems* (Paris, France) (CHI EA '13). Association for Computing Machinery, New York, NY, USA, 2351–2354. <https://doi.org/10.1145/2468356.2468767>
- [35] Matthias Söllner, Axel Hoffmann, Holger Hoffmann, Arno Wacker, and Jan Marco Leimeister. 2014. Understanding the formation of trust. In *Socio-technical design of ubiquitous computing systems*. Springer, 39–58.
- [36] Maarten Ter Huurne, Amber Ronteltap, Rense Corten, and Vincent Buskens. 2017. Antecedents of trust in the sharing economy: A systematic review. *Journal of Consumer Behaviour* 16, 6 (2017), 485–498.
- [37] Niels van Doorn. 2017. Platform labor: on the gendered and racialized exploitation of low-income service work in the 'on-demand' economy. *Information, Communication & Society* 20, 6 (jun 2017), 898–914. <https://doi.org/10.1080/1369118X.2017.1294194>
- [38] David Wachsmuth and Alexander Weisler. 2018. Airbnb and the rent gap: Gentrification through the sharing economy. *Environment and Planning A: Economy and Space* 50, 6 (2018), 1147–1170. <https://doi.org/10.1177/0308518X18778038>
- [39] Jeannette M. Wing. 2021. Trustworthy AI. *Commun. ACM* 64, 10 (sep 2021), 64–71. <https://doi.org/10.1145/3448248>
- [40] Liudmila Zavolokina, Noah Zani, and Gerhard Schwabe. 2020. Designing for trust in blockchain platforms. *IEEE Transactions on Engineering Management* (2020).
- [41] Jiang Zhu and Olivera Marjanovic. 2021. A Different Kind of Sharing Economy: A Literature Review of Platform Cooperatives. In *Hawaii International Conference on System Sciences 2021 (HICSS-54)*. 4128–4137. https://aisel.aisnet.org/hicss-54/in/crowd-based_platforms/2